

CustomVRAD

Technical Report: Lightmap Baking Algorithms

Path-traced and radiosity pipelines for Source SDK 2013 / Garry's Mod

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Abstract

CustomVRAD is a 64-bit lightmap compiler derived from Valve VRAD (Source SDK 2013) for Garry's Mod. It retains the classical patch-radiosity pipeline and optionally replaces world-face (and static-prop) lighting with a DirectX Raytracing (DXR) path tracer. Extensions include hero-wavelength spectral transport, contact-hardening soft shadows, textured emissive and filter materials, spatial volume overrides, multi-scale ambient occlusion, IESNA photometric spot evaluation, directional projected-texture modulation on `light_spot`, physically based underwater media (Beer-Lambert / Kd and homogeneous volume path tracing), BSP-shipped volumetric fog grids with convex brush SDFs and a screenspace shader, GPU texture shadows for `$alphatest/$translucent` props, and lightmap post-processes (denoising, seam stitching, edge pull). This document states the algorithms as implemented: estimators, coordinate conventions, parameter semantics, and the relationship between classic and path-traced bake modes.

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1 System overview

CustomVRAD produces HDR and LDR lightmaps compatible with the Source engine. World-face lighting uses exactly one of two pipelines:

1. **Classic radiosity** — stock VRAD BuildFacelights and iterative patch gather, optionally accelerated by OpenCL (`-gpu`).
2. **Path tracing** — D3D12 DXR baker (`-pathtrace` / `-dxr`). On success it replaces world-face direct lighting and radiosity bounce. Static props may share the integrator under `-StaticPropLighting`.

Leaf ambient cubes, detail props, and engine-exported worldlights remain on stock paths, with volume-aware CustomVRAD extensions where noted. If DXR initialisation or bake fails, the compiler falls back to classic radiosity for world faces (and CPU gather for props).

GPU backends

`-pt_gpu` selects the DXR RayQuery luxel baker. `-gpu` selects an independent OpenCL path for classic radiosity gather and batched AO occlusion. The two backends are not interchangeable.

1.1 Data flow

1. Load BSP, materials (VMT/VTF), entities, static props.
2. Build ray-tracing acceleration (CPU KD-tree and/or DXR TLAS/BLAS).
3. Direct lighting (classic) *or* PathLi bake (pathtrace).
4. Radiosity bounce (classic only, if `-bounce N > 0`).
5. Optional AO, denoise, seam stitch, edge-pull-aware sampling.
6. Write lightmaps / prop vertex colours into the BSP.

1.2 Notation

Symbol	Meaning
$\mathbf{x}, \mathbf{n}$	Sample position and shading normal
ρ	Diffuse albedo (linear RGB or spectral reflectance)
β	Path throughput
L_e	Emitted radiance
V	Binary (or soft) visibility
N	<code>-pt_bounces</code> — number of <i>indirect</i> hops
M	Path depth = $N + 1$ (primary vertex + N bounces)

2 Path tracing (DXR)

The path tracer is a unidirectional Lambertian integrator with next-event estimation (NEE) at every path vertex. It is implemented twice:

- **GPU** (`-pt_gpu`, default): compute shader using DXR 1.1 RayQuery, TLAS over world + optional prop BLASes (`pathtrace_dxr_gpu_bake.inl`).

- **CPU** (`-pt_cpu`): threaded SSE reference (`PtPathLi` in `pathtrace_dxr.cpp`). RGB-only; no spectral.

2.1 PathLi estimator

At luxel (or prop sample) $(\mathbf{x}_0, \mathbf{n}_0)$ the estimator is

$$\hat{L} = \sum_{k=0}^m \beta_k L_{\text{NEE}}(\mathbf{x}_k, \mathbf{n}_k) + \beta_{m'} L_{\text{sky}}(\mathbf{x}_{m'}), \quad (1)$$

where m is the last surface vertex before termination, and the sky term is added when a BSDF sample escapes to the sky (ambient sky is *not* NEE-sampled). Under `-uw_volume`, medium free-flight events may intervene between surface vertices (section 8).

BSDF. Surfaces are Lambertian,

$$f_r(\omega_i, \omega_o) = \frac{\rho}{\pi}, \quad p_{\text{bsdf}}(\omega) = \frac{\cos \theta}{\pi}. \quad (2)$$

After a non-sky hit the throughput update simplifies to

$$\beta_{k+1} = \beta_k \rho_k \quad (\text{albedo clamped channelwise to } [0, 1]). \quad (3)$$

Russian roulette. Starting at bounce index $k \geq 1$,

$$q = \min(0.95, \max(\beta_r, \beta_g, \beta_b)) \quad (\text{spectral: mean of four } \beta_\lambda). \quad (4)$$

Terminate if $u > q$; otherwise $\beta \leftarrow \beta/q$.

2.2 Bounce semantics

$\text{pathDepth} = N + 1, \quad N = \text{-pt_bounces}$

N counts **indirect hops after the luxel**. Vertex $k = 0$ is always the receiving sample (primary NEE = direct lighting). Therefore:

N	Path depth	Effect
0	1	Direct lights + sky escape; no GI
1	2	One bounce of reflected light
6	7	Typical indoor quality (config sheet)

Defaults: $N = 3$ when unset; $N = 1$ under `-fast`; clamp $[0, 16]$. Unset is distinguished from explicit zero by an internal sentinel (-1). `-pt_prop_bounces` mirrors this for vertex prop quality (auto = world N).

2.3 Next-event estimation

At every vertex the integrator evaluates (or samples) lights:

- **Sun / sky disc** — always evaluated; Veach power-heuristic MIS vs. cosine BSDF for the angular sun.

- **Point / spot** — all lights when `-pt_lights 0`; otherwise power-proportional CDF sample of K locals with weight $1/(p_i K)$. Bounce 0 always evaluates all locals on GPU.
- **\$vrad_emit_area_triangles** — power-weighted triangle selection + uniform area sample (PBRT-style).
- **Legacy emit_surface** — NEE-only proxies (no BSDF hit contribution; avoids double-counting).

Sun MIS. For sun solid angle $\Omega = 2\pi(1 - \cos \theta_{\max})$,

$$p_{\text{light}} = \frac{1}{\Omega}, \quad w_{\text{light}} = \frac{p_{\text{light}}^2}{p_{\text{light}}^2 + p_{\text{bsdf}}^2}. \quad (5)$$

Point / spot attenuation.

$$A(d) = \frac{1}{c + \ell d + q d^2}, \quad (6)$$

with optional hard distance cap and quintic smootherstep fade $s(t) = t^3(6t^2 - 15t + 10)$.

Area emission contribution (triangle sample with density p_A):

$$L_{\text{area}} = L_e \frac{(\mathbf{n}_r \cdot \boldsymbol{\omega})(-\mathbf{n}_e \cdot \boldsymbol{\omega}) V}{d^2 p_A}. \quad (7)$$

2.4 Soft shadows and contact hardening

Local lights may use a soft disk of radius r (`-pt_lightradius`, or entity `light_volume Radius`). The sun uses an angular cap (`-softsun` / entity `SunSpreadAngle`).

Adaptive sample count (CHSS-style) from a probe blocker distance t_{hit} and light distance d :

$$w_{\text{pen}} = \frac{t_{\text{hit}} r s_p}{\max(d - t_{\text{hit}}, 10^{-3})}, \quad n = \text{clamp}(1 + \lfloor 0.35 w_{\text{pen}} + 0.5 \rfloor, 1, n_{\max}), \quad (8)$$

where $s_p = \text{-pt_lightpenumbra}$ (default 1) and $n_{\max} = \text{-pt_softsamples}$ (default 16, clamp 1–64). Unoccluded lights use a cheap angular estimate, capped at four rays.

`-pt_softmode direct` (default) enables soft sampling only at the luxel; `all` enables it at every path vertex.

2.5 Firefly clamp

Hue-preserving max-channel clamp with cap $C_{\max} = 2500$:

$$C' = \begin{cases} C \cdot C_{\max} / \max(C) & \text{if } \max(C) > C_{\max}, \\ C & \text{otherwise.} \end{cases} \quad (9)$$

2.6 Sampling and luxel AA

- **-pt_samples S** : spp per luxel (quality-derived if unset; clamp 1–4096). GPU uses fixed S ; CPU may adaptively raise spp (up to $4\times$, cap 64) when coefficient of variation is high.
- **-pt_aa A** : deterministic $A \times A$ footprint grid over the luxel ($A \in [1, 5]$, default 3). Spatial AA, not a multiplicative spp loop.

2.7 Post-bake lightmap cleanup

After PathLi accumulation the baker:

1. Fills chart holes / invalid samples.
2. Dilates suspicious dark edge outliers (sample-in-solid).
3. Optionally denoises (section 12).
4. Always applies two FXAA-like 3×3 high-contrast evening passes (blend starts near 25% relative luminance jump, full near 55%, neighbour blend capped at 65%).

3 Spectral transport

Default on GPU: `-pt_spectral`. Disable with `-pt_nospectral`. CPU PathLi remains linear RGB.

3.1 Hero wavelengths

Four stratified wavelengths per path over [380, 780] nm:

$$\lambda_i = 380 + (i + \xi_i) \frac{400}{4}, \quad i = 0, 1, 2, 3. \quad (10)$$

3.2 RGB $\leftrightarrow$ spectrum

- **Upsampling:** Smits (1999) bases as tabulated in PBRT-v3 (32 samples over the visible range) — separate bases for reflectance and illuminants.
- **Transport:** illumination adds and reflectance multiplies in λ -space.
- **Projection:** Wyman analytic CIE 1931 2° CMFs $\rightarrow$ XYZ, then IEC 61966-2-1 / Rec. 709 matrix to linear sRGB.

Monte Carlo reconstruction weight uses the CIE Y integral $Y_{\text{int}} = 106.856895$:

$$(X, Y, Z) += \frac{L(\lambda)}{p_\lambda Y_{\text{int}}} (\bar{x}(\lambda), \bar{y}(\lambda), \bar{z}(\lambda)). \quad (11)$$

Spectral mode is implemented in the GPU HLSL baker only. `-pt_cpu` ignores `-pt_spectral` and transports in linear RGB.

4 Classic radiosity

Used when pathtrace is off or fails. `-bounce 0` skips the entire bounce stage (direct only) — the classic analogue of `-pt_bounces 0`.

4.1 Patches and form factors

Faces are subdivided into patches (unless bounce count is zero). Visibility is PVS-pruned and ray-tested. Far-field differential form factor:

$$F_{1\leftrightarrow 2} = \frac{(-\hat{d} \cdot \mathbf{n}_1)(\hat{d} \cdot \mathbf{n}_2)}{\|\mathbf{x}_1 - \mathbf{x}_2\|^2}. \quad (12)$$

Polygon-to-differential form factors are used when patch size invalidates the far-field approximation. Raw transfer = source area $\times$ form factor. Receiver lists are normalised:

$$s = \begin{cases} 1/T & T > \pi, \\ 1/\pi & T \leq \pi, \end{cases} \quad T = \sum_j t_j. \quad (13)$$

4.2 Gather iteration

Emitted light E_j and patch reflectance ρ_j gather as

$$A_i = \sum_j E_j \odot \rho_j t_{ij}. \quad (14)$$

Parent patches are area-weighted averages of children. Iteration stops after **-bounce** rounds (default 100) or when added energy falls below $\max(1, 0.0015 E_{\text{bounce } 1})$.

4.3 Textured bounce (**-texbounce**)

Samples **\$basetexture** at each leaf-patch origin for colour bleed. Reflectance is clamped to $[0, 0.99]$. Supports VTF 7.5; falls back to flat texdata average.

4.4 Cavity-damped energy (**-energy** / **-cavity**)

Opt-in. Enclosure estimate $e = \min(1, T/\pi)$ yields

$$d_E = 1 - e^2(1 - C), \quad d = 1 + (d_E - 1)\alpha, \quad (15)$$

where $C \in [0.25, 1]$ is the fully enclosed scale (default 0.70) and α is the global or volume **EnergyMode** blend. Open areas stay near stock; closed rooms recirculate less.

4.5 Artistic bounce controls

- **-bounce_boost** B : scales *integrated bounce only* after all iterations ($B \in [0, 16]$; direct unchanged).
- **-bounce_chroma** C : Oklab saturation of gathered bounce, strongest on early bounces:

$$(a, b) \leftarrow (a, b) \left(1 + \frac{C}{1+k}\right), \quad k = 0\text{-based bounce index}, \quad (16)$$

keeping perceptual lightness L . Requires **-texbounce**.

Pathtrace ignores **-bounce_chroma** (non-physical). Bounce volumes may still apply boost to pathtrace albedo/throughput.

4.6 Radial splat and bounce weld

Bounce is written to luxels with radial kernel radius scaled by **-bounce_soft** (default 1, range 0.5–4).

-bounce_weld (off by default) restricts coplanar neighbour-face GI to a shared-edge band (normal $\cdot \geq 0.985$; falloff beyond ~ 24 patch radii). Reduces rectangular GI smear across coplanar V BSP splits; can blotch ceilings.

4.7 OpenCL gather (-gpu)

Median-split BVH (≤ 8 tris/leaf). Kernels batch visibility and non-bump bounce gather over a CSR-like transfer graph (same $E_j \rho_j t_{ij}$ math). Bump gathers stay on CPU. Default batch 32768 rays; errors fall back to CPU.

5 Materials

5.1 Emissive surfaces (\$vrad_emit*)

Enabled by `$vrad_emit 1` or `$vrad_emitstrength > 0` (default strength 200 when flag-only).

Colour from `$vrad_emitmap/$vrad_emissivemap`, else `$basetexture`; optional grayscale `$vrad_emitmask`.

Texture RGB is linearised approximately as $x^{2.2}$, then scaled by strength and `lightscale`. Pathtrace builds a fan-triangulated mesh and uses area NEE; **-pt_emit_samples 0** evaluates all triangles (default 64 power samples). Classic mode aggregates to a softened area proxy plus a later self-illumination pass.

5.2 Colored glass filters (\$vrad_filter*)

Pathtrace-only thin-sheet transmission. Ordinary `$translucent` glass does *not* tint light unless `$vrad_filter 1`.

Transmittance construction.

1. Sample filter map (or basetexture); square RGB for linearisation.
2. Thickness: $T \leftarrow T^{\text{thickness}}$ (Beer-like artistic).
3. Strength: Oklab lerp white $\rightarrow$ filter.
4. Opacity: multiply by $(1 - \text{opacity})$.

Rays may traverse up to 64 filter hits, deduplicating brush-sheet entrance/exit pairs within 12 world units. Default two-sided; `$vrad_filter_onesided 1` = front only.

Stacked panes (Oklab). Energy transport stays linear; stacking is perceptual:

$$L' = L_0 L_1, \quad a' = a_0 + a_1, \quad b' = b_0 + b_1. \quad (17)$$

5.3 Texture shadows (-textureshadows)

Alpha-masked static props cast coverage-aware shadows without requiring the MDL `STUDIOHDR_FLAGS_CAST_TEXTURE` bit when **-textureshadows** is set. Materials with `$alphatest` or `$translucent` (non-zero) contribute basetexture alpha into a GPU atlas and per-triangle UV meta.

Under **-pathtrace** / DXR RayQuery:

- **\$alphatest** — stochastic any-hit cutout (Enderton-style): a uniform $U \sim \mathcal{U}[0, 1]$ rejects the hit when $U \geq \alpha$.
- **\$translucent** — expected coverage transmittance $T \leftarrow T(1 - \alpha)$ along VisRGB / SunVisRGB (soft dappled shadows). Binary occlusion queries still use stochastic testing.

Foliage triangles remain shadow casters only (skipped as GI bounce surfaces). Classic SSE pathtrace-off continues to use stock CPU coverage accumulation.

6 Oklab perceptual ops

Björn Ottosson’s Oklab is used wherever artistic colour mixing must stay perceptually coherent. Linear sRGB $\leftrightarrow$ Oklab follows the standard LMS cube-root pipeline (coefficients in `oklab.h`).

Use site	Operation
Filter strength	Oklab lerp(white, T , strength)
Stacked filters	L multiply, a, b add
-bounce_chroma	Scale a, b ; keep L
Env-vol inbound tint	Keep light L ; take tint a, b

NEE intensities and $\rho \times L$ products remain in linear RGB (or λ).

7 Volume entities

All brush volumes share soft AABB membership: soft-min signed distance, quintic smootherstep blend shell (`BlendDistance` / `BlendMode`), and priority resolution (higher wins; ties $\rightarrow$ smaller AABB). Neighbour volumes use a soft Voronoi split so outside halos do not tint each other.

7.1 light_env_vol

Local sky / sun / ambient override. Volume lights are bake-only (not exported as engine worldlights). Leaf ambient cubes accumulate volume-blended sun when samples see the sun through sky. Shadow filters use the hard AABB via `OutsideCastShadowIn` / `InsideCastShadowOut`.

Optional `BounceVolColor` / `BounceVolBright` recolour bounced light inside the core to the volume’s `_light` hue.

7.2 light_absorb

Direct multiplier $1 - w$ when `AbsorbDirect`; bounce multiplier $1 - \max(w_e, w_r)$ when `AbsorbBounce` (default on; strength default 0.5).

7.3 light_bounce_vol

Local overrides for `BounceBoost`, `BounceChroma`, and `EnergyMode`, soft-blended with CLI globals.

7.4 light_volume

Soft sphere point light: low-discrepancy samples inside radius R (default 16). Sample count scales with R (~ 16 at default, cap 40; reduced with **-fast**). Early-out when fully lit/shadowed. Exported as a hard point light at the centre (softness is bake-only).

7.5 fog_volume (runtime volumetric fog)

Bake-time construction of a 3D irradiance grid and packing of a screenspace fog draw into the BSP (no addon). Let σ be optical density per world unit, g Henyey–Greenstein anisotropy, and c fog tint. Raymarching uses an AABB expanded by `BlendDistance` under Outside/Center blend modes for the light grid / early-out; density membership uses a **convex brush SDF** baked into atlas alpha from the entity’s planes (vertex-edited / clipped brushes, not only axis-aligned boxes). Grid resolution is `GridSpacing` clamped so that each axis has at most 64 samples (64^3 cap). Optional FBM noise (`NoiseScale`, `NoiseCoverage`) modulates density; wind scrolls the atlas in world space.

`fog_volume_blocker` carves fog (soft blend, also plane-aware) into atlas alpha and is rewritten to `info_null`. Enabled volumes are rewritten to `lua_run_on_client` stubs that hook `RenderScreenspaceEffects`. Clients that block map client Lua will not draw fog.

7.6 water_light_vol

Optional override AABB for underwater IOP fitting (section 8): Jerlov class, σ_a , σ_s , HG g , fog colour / end. Priority resolves overlapping overrides.

8 Underwater media

Physically motivated participating-medium bake for BSP water (`leafWaterData` / `CONTENTS_WATER`). Units: hammer inches; ocean tables in 1/m are scaled by 1/39.3700787.

8.1 Modes

Flag	Mode	Estimator
<code>-underwater</code>	A	Beer–Lambert segment T + Kd downwelling for sky/sun
<code>-uw_volume</code>	C	Homogeneous free-flight volume PT (implies A media)
<code>-uw_off</code>	—	Force disable
<code>-uw_maxscatter</code> N	C cap	Max medium events (default 8, max 64)

Both CPU PathLi and the GPU RayQuery baker implement the modes; Mode A also scales classic `GatherSampleLight` for direct sun/sky/local paths.

8.2 IOP construction from VMT fog

Source `$fogend` is authored for *screenspace fog*, not lightmap optical depth. Fit:

$$\sigma_{\text{lum}} = \frac{-\ln 0.05}{\text{fogend}} \cdot s_{\text{bake}}, \quad s_{\text{bake}} = 0.28, \quad (18)$$

channel-biased by inverse fog colour so transport shifts toward the fog tint. Scatter fraction 0.45 of extinction; diffuse downwelling

$$k_d = 0.35 \sigma_t \quad (19)$$

($K_d \ll$ beam extinction). Optional `$vrad_uw_jerlov`, `$vrad_uw_sigma_a`/`$vrad_uw_sigma_s`, `$vrad_uw_g`, or `water_light_vol` overrides. Default fog end when missing: 800.

8.3 Mode A (attenuation)

Segment transmittance through each water AABB of length L :

$$T = \exp(-\sigma_t L) \quad (\text{componentwise}). \quad (20)$$

Sky/sun irradiance at a submerged receiver uses *only*

$$S_{\text{sky}}(\mathbf{x}) = \exp(-\mathbf{k}_d z_{\text{depth}}), \quad (21)$$

floored at 0.04 per channel — not $T \times S_{\text{sky}}$ (double counting). Local lights use segment T only. Mode C disables Kd.

8.4 Mode C (volume path tracing)

Inside water, free-flight distance $t = -\ln(1 - u) / \max(\sigma_t)$ is accepted only if t is strictly less than the AABB exit distance; otherwise the ray continues as a surface path (no fake boundary scatter). On a medium event, throughput is multiplied by T to the event and by scatter albedo σ_s / σ_t , HG direction is resampled, and a small sky-veil term reinjects ambient fill. Escapes to sky apply segment T through the water column (Kd off).

Dense Source water materials with short `$fogend` still darken deep pools; author longer fog ends, Jerlov classes, or `water_light_vol` IOPs for clear water.

9 Ambient occlusion

Multi-scale hemisphere AO in `FinalLightFace`, enabled by `-ao` / `-ao_*` or entities `light_ao` / `light_ao_vol`.

$$AO(\mathbf{x}) = \frac{\sum_i w_i V_i}{\sum_i w_i}, \quad w_i = \max(\mathbf{n} \cdot \omega_i, 0.08). \quad (22)$$

Directions with $\mathbf{n} \cdot \omega < -0.25$ are rejected (small wrapped hemisphere for contact darkening). Ray origin:

$$\mathbf{x}_0 = \mathbf{x} + \mathbf{n} b + \omega \max(0.05, 0.5b). \quad (23)$$

Final multiplier $M = \max(0, 1 - S + S \cdot AO)$ with strength $S \in [0, 8]$.

Defaults: samples 16, distance 48, strength 1, bias 0.25. Optional bilateral luxel denoise:

$$w = \exp\left[-\frac{\|\Delta x\|^2}{2\sigma_s^2} - \frac{(AO_j - AO_i)^2}{2(0.12)^2}\right]. \quad (24)$$

With `-gpu`, AO can use batched OpenCL occlusion.

Under `-pathtrace` with $N = \text{pt_bounces} > 0$, post-multiply AO is *skipped*: the path tracer already accounts for geometric occlusion. `-ao_force` re-enables the multiply for artistic double-darkening. With $N = 0$ (direct+sky only), AO still applies as a stand-in for missing GI.

10 Soft sun

Replaces stock's fixed 30 random cone rays:

- Low-discrepancy golden-angle spiral in the sun cone.
- Adaptive count ~ 9 at 2° up to ~ 40 at 50° .
- Early-out after a short probe when fully lit or fully shadowed.

0° = hard sun. Typical soft look: 0.5° – 3° .

11 Photometric spots and projected textures

Bake-only extensions on stock `light_spot`. Engine-exported worldlights retain cone keys for dynamic lighting; lightmap evaluation may substitute an IESNA intensity table and/or a projected Valve Texture Format (VTF) multiplier.

11.1 IESNA photometry

When the entity key `IES` names an LM-63 file under `garrysmo/IES/` (or a per-map subdirectory `IES/maps/mapname/`), the angular candela table replaces `_inner_cone`, `_cone`, and `_exponent` for the bake. Only `TILT=NONE` profiles are supported. Distance attenuation (`_constant_attn`, ...) is unchanged.

Let $\mathbf{b}$ be the beam axis and $\mathbf{r}$ a right vector from entity angles. For a unit direction $\boldsymbol{\omega}$ from light to sample, vertical and horizontal angles (θ, ϕ) index a peak-normalized table $I(\theta, \phi) \in [0, 1]$. The photometric weight is $w_{\text{IES}} = s I(\theta, \phi)$ with scale $s = \text{IESScale}$ (default 1). Optional `IESBrightness` ≥ 0 overrides stock `_light` brightness for the bake only; `IESMaxIntensity` caps per-sample RGB after attenuation.

11.2 Projected texture classification

The key `ProjectedTexture` names a material under `materials/` (optional `materials/` prefix and `.vtf` suffix are stripped). The loader classifies the VTF:

Kind	Detection	Domain
Planar (2D)	No cubemap / no usable six-face layout	Spot cone
Cubemap	Six faces (<code>IsCubeMap</code> / <code>ENVMAP</code> + face count)	Full sphere
Spherical 2D	<code>ENVMAP</code> without six faces	Full sphere

Spherical 2D maps use equirectangular sampling when width $\geq \frac{3}{2}$ height; otherwise an OpenGL-style spheremap / matcap projection. Cubemap VTFs (including VTF 7.5 with CRC resources) are decoded to RGBA face arrays; GPU upload stores six consecutive `Texture2DArray` layers per cubemap in Source `CUBEMAP_FACE_*` order ($+X, -X, +Y, -Y, +Z, -Z$).

11.3 Planar framing in the outer cone

For planar projections, let $\alpha = \arccos(\text{cone})$ (outer half-angle) and $\boldsymbol{\omega}$ the light-to-sample direction. With $\cos \theta = \boldsymbol{\omega} \cdot \mathbf{b}$:

$$u' = \frac{\boldsymbol{\omega} \cdot \mathbf{r}}{\cos \theta \tan \alpha}, \quad v' = \frac{\boldsymbol{\omega} \cdot \mathbf{u}}{\cos \theta \tan \alpha}, \quad (25)$$

where $\mathbf{u}$ completes a right-handed basis. Texture coordinates are

$$(u, v) = (\frac{1}{2} + \frac{1}{2} f u', \frac{1}{2} + \frac{1}{2} f v'), \quad (26)$$

with framing factor $f = 1$ for `ProjectedTextureMode=fill` (square sides on the cone; corners clipped) and $f = \sqrt{2}$ for `fit` (square inscribed in the cone). Samples with $(u, v) \notin [0, 1]^2$ contribute zero (hard mask). Cubemap and spherical modes ignore the cone and evaluate over 4π .

11.4 Cubemap direction mapping

Entity angles define a light-local frame $(\mathbf{r}, \mathbf{u}, \mathbf{b})$ (right, up, beam). A world direction $\boldsymbol{\omega}$ maps to Source cube axes (Z-up) by

$$\mathbf{d} = (\boldsymbol{\omega} \cdot \mathbf{r}, \boldsymbol{\omega} \cdot \mathbf{b}, \boldsymbol{\omega} \cdot \mathbf{u}), \quad (27)$$

i.e. $(d_x, d_y, d_z) = (\text{right}, \text{forward}, \text{up})$. Face selection and face coordinates follow the Direct3D cubemap table (identical to hardware `texCUBE` / Source VTF layout):

$$\begin{aligned} +X : sc &= -d_z, tc = -d_y, & -X : sc &= +d_z, tc = -d_y, \\ +Y : sc &= +d_x, tc = +d_z, & -Y : sc &= +d_x, tc = -d_z, \\ +Z : sc &= +d_x, tc = -d_y, & -Z : sc &= -d_x, tc = -d_y, \end{aligned} \quad (28)$$

with major-axis magnitude m and

$$u = \frac{1}{2}(sc/m + 1), \quad v = \frac{1}{2}(tc/m + 1). \quad (29)$$

Faces clamp at edges. CPU and GPU bakers share this convention.

11.5 Composite intensity

Let L_0 be the attenuated spot intensity (photometric or cone), $\mathbf{P} \in [0, 1]^3$ the projected RGB sample, and w_{IES} the optional IES weight. The bake contribution uses

$$L = L_0 w_{\text{IES}} \mathbf{P} \quad (30)$$

(componentwise; missing IES or projection $\Rightarrow$ factor 1). When both IES and `ProjectedTexture` are set, IES acts solely as an angular mask on the projection. Evaluation is wired through classic gather, CPU PathLi, GPU PathLi, and static-prop lighting.

12 Denoising

Opt-in via `-pt_denoise`; select backend with `-pt_denoiser`.

12.1 OIDN (default)

Intel Open Image Denoise RTLightmap model on CPU. One thread-local filter per bake worker (pinned to avoid oversubscription). Charts smaller than 16^2 are edge-clamp padded.

12.2 OptiX

NVIDIA CUDA HDR denoiser (no albedo/normal guides). Shared serialised GPU resources; same 16^2 padding.

12.3 Sakai (statistical)

Sakai et al. (2024)–inspired Welch-gated repeated 5×5 filter using variance of the mean where available:

$$t = \frac{|L_i - L_j|}{\sqrt{\text{Var}(\bar{L}_i) + \text{Var}(\bar{L}_j) + (0.04L_{\text{ref}})^2 + \varepsilon}}. \quad (31)$$

If $t > 1.35$, neighbour weight is attenuated by $\exp[-1.5((t - 1.35)/1.35)^2]$. Runs $1 + \text{radius}$ passes, then two low-gradient 3×3 evening passes. `-pt_denoise_radius` / `-pt_denoise_strength` apply here (OIDN/OptiX largely ignore radius).

13 Static prop lighting

13.1 Classic speedups

Four-wide SSE direct gather; with `-gpu`, bounce rays that miss / hit sky skip the BSP lightmap walk.

13.2 Pathtrace props

Under `-pathtrace` + `-StaticPropLighting`:

- **Prop lightmap texels** — same spp / bounces / NEE as world.
- **Vertex lighting** — with `-pt_prop_vertgrid` $G > 0$ (default 4): each LOD0 triangle gets an 8×8 chart packed into a 64×64 -cell atlas page (512×512), inset by 1 texel (right-triangle lattice in the 6×6 content region, ≈ 21 samples/tri). Pages are hole-filled and, when `-pt_denoise` is on, denoised with the world-face backends. Each vertex takes the closest valid sample from every adjacent triangle chart and averages those colours. Unlit verts inherit from nearest lit. $G = 0$ = old one-sample-per-vertex.

Override quality with `-pt_prop_samples` / `-pt_prop_bounces` (defaults: $\max(8, S/4)$ spp; bounces = world). Multi-LOD models share lit colours across LODs.

14 Lightmap sample quality

14.1 Edge pull

Default on (inset 0.5 luxels; range 0–2 via `-edgepull`). For chart width W , sample coordinate s is clamped to $[i, W - 1 - i]$ (similarly t). Moves edge samples off adjacent solids — thin walls, jambs, floor/wall scallops. `-noedgepull` restores stock positions.

14.2 Seam stitching

On by default after `FinalLightFace`. Coplanar faces that share an edge (VBSP splits) blend near-edge luxels toward the neighbour value at the same world position so both sides converge to a common average. Per lightstyle and bump layer; displacements skipped. Disable with `-nostitch`.

15 Reference integrator (pseudocode)

Algorithm 1 PathLi — one spp at a surface sample

Require: position $\mathbf{x}$, normal $\mathbf{n}$, bounce limit N , seed

```
1:  $\beta \leftarrow \mathbf{1}$ ;  $L \leftarrow \mathbf{0}$ ; depth  $\leftarrow N + 1$ 
2: for  $k \leftarrow 0$  to depth  $- 1$  do
3:    $L \leftarrow L + \beta \cdot \text{NEE}(\mathbf{x}, \mathbf{n}, k)$  ▷ soft only if  $k = 0$  or softmode=all
4:    $\omega \leftarrow \text{CosineHemisphere}(\mathbf{n})$ 
5:   trace closest hit along  $\omega$ , applying filter panes to  $\beta$ 
6:   if miss or sky then
7:      $L \leftarrow L + \beta \cdot L_{\text{sky}}(\mathbf{x})$ ; break
8:   end if
9:    $\beta \leftarrow \beta \cdot \rho_{\text{hit}}$  ▷ or spectral Reflect( $\lambda$ )
10:  if  $k \geq 1$  then
11:    Russian roulette on  $\beta$ 
12:  end if
13:   $\mathbf{x}, \mathbf{n} \leftarrow$  hit point / oriented normal
14: end for
15: return firefly-clamp( $L$ )
```

16 Parameter cheat sheet

Flag	Default	Role
<code>-pathtrace</code>	off	Enable DXR baker
<code>-pt_gpu/-pt_cpu</code>	GPU	Integrator backend
<code>-pt_samples <i>S</i></code>	quality	spp per luxel
<code>-pt_bounces <i>N</i></code>	3 (1 fast)	Indirect hops; 0=direct+sky
<code>-pt_aa <i>A</i></code>	3	Luxel footprint grid
<code>-pt_spectral</code>	on	Hero-λ GPU transport
<code>-pt_nospectral</code>	—	Linear RGB path transport
<code>-pt_lights <i>K</i></code>	0=all	Local NEE samples (bounce> 0)
<code>-pt_emit_samples</code>	64	Area-emit NEE draws (0=all)
<code>-pt_lightradius</code>	0	Soft disk for point/spot
<code>-pt_lightpenumbra</code>	1	CHSS scale
<code>-pt_softsamples</code>	16	Soft ray cap
<code>-pt_softmode</code>	direct	Soft at luxel / all vertices
<code>-pt_denoise</code>	off	Enable denoise
<code>-pt_denoiser</code>	oidn	oidn optix sakai
<code>-pt_prop_vertgrid</code>	4	Prop vert atlas on (0=per-vert)
<code>-underwater</code>	off	Mode A Beer-Lambert + Kd
<code>-uw_volume</code>	off	Mode C free-flight volume PT
<code>-uw_maxscatter <i>N</i></code>	8	Cap medium scatters (Mode C)
<code>-bounce <i>N</i></code>	100	Classic radiosity rounds (0=direct)
<code>-texbounce</code>	off	Textured radiosity albedo
<code>-energy/-cavity</code>	off / 0.70	Cavity damp
<code>-bounce_boost</code>	1	Artistic bounce scale
<code>-bounce_chroma</code>	0	Early-bounce Oklab sat
<code>-bounce_weld</code>	off	Coplanar edge GI weld
<code>-gpu</code>	off	OpenCL classic gather
<code>-textureshadows</code>	off	Prop α shadows (<code>\$alphatest</code> / <code>\$translucent</code>)
<code>-ao</code>	off	Hemisphere AO pass
<code>-ao_force</code>	off	AO even under pathtrace GI
<code>-edgepull</code>	0.5	Luxel edge inset
<code>-nostitch</code>	—	Disable seam stitch

Config presets

Hammer-friendly flag sheets live under `configs/` and load via `-config full` (searches `.cfg/.txt` next to the binary). Do not put `-game` / map path inside the config file.

17 Implementation map

Module	Responsibility
<code>pathtrace_dxr.cpp</code>	CPU PathLi, bake orchestration, AA, cleanup
<code>pathtrace_dxr_gpu_bake.inl</code>	HLSL PathLi (spectral, NEE, filters)
<code>pathtrace_dxr_device.*</code>	D3D12 / DXR device, AS, dispatches
<code>pt_spectral.h</code>	Smits bases, CIE CMFs, RGB $\leftrightarrow\lambda$
<code>oklab.h</code>	Perceptual colour ops
<code>vrad_emit*.cpp</code>	Textured emission + area mesh
<code>vrad_filter.cpp</code>	Colored glass transmittance
<code>vrad.cpp</code> / <code>vismat.cpp</code>	Classic radiosity loop
<code>radial.cpp</code>	Bounce splat / weld
<code>vrad_gpu.*</code>	OpenCL BVH + gather
<code>ao.cpp</code>	Multi-scale AO
<code>absorb.cpp</code> / <code>bounce_vol.cpp</code>	Volume effects
<code>fog_volume.*</code>	Fog grid bake + BSP Lua/shader pack
<code>water_medium.*</code>	Underwater IOPs, Beer–Lambert, volume PT hooks
<code>ies_profile.*</code>	IESNA LM-63 load + angular eval
<code>light_projection.*</code>	Projected VTF load, CPU sample, GPU arrays
<code>pathtrace_*denoise*</code>	OIDN / OptiX / Sakai
<code>vradstaticprops.cpp</code>	Prop SSE + pathtrace vert atlas
<code>lightmap.cpp</code>	Edge pull, final face lighting, spot parse
<code>loadcmdline.cpp</code>	-config preset tokenization

References & provenance

- Kajiya, J. T. — The rendering equation (SIGGRAPH 1986).
- Veach, E. — Robust Monte Carlo methods for light transport (PhD, 1997): power-heuristic MIS.
- Smits, B. — An RGB to spectrum conversion for reflectances (JGTools 1999); PBRT-v3 tabulated bases.
- Wyman, C. et al. — Analytic CIE 1931 CMF fits.
- Ottosson, B. — Oklab (2020).
- Pharr, Jakob, Humphreys — *Physically Based Rendering* (area lights, spectra).
- Sakai et al. — Statistical lightmap filtering (2024 inspiration for the Sakai denoiser).
- Intel Open Image Denoise — **RTL**ightmap model.
- NVIDIA OptiX — HDR AI denoiser.
- IESNA LM-63 — standard file format for luminous intensity distributions.
- Microsoft Direct3D — cubemap face selection and (s, t) mapping (**tex**CUBE / environment maps).
- Jerlov / Solonenko & Mobley — ocean optical water types (IOP tables).
- Jaffe–McGlamery — classical underwater image formation (direct / forward / backscatter decomposition).
- Monzon et al. (2024) — spectral underwater radiative transfer (downwelling K_d).

- Henyey & Greenstein — phase function for volume scattering.
- Valve Source SDK 2013 — stock VRAD radiosity / lightmap pipeline; VTF cubemap face order (`CUBEMAP_FACE_*`).

Algorithms are described as implemented in the CustomVRAD source tree. CLI defaults and clamps may change between revisions; `vrad.exe help` output and `configs/full.cfg` define shipping presets.